

Simulation Model & Environment Standard - (SMES)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Operational Layer: Simulation Model & Environment Governance Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Parent Standard

Governed Space: Simulation Model & Environment

Version: 1.0

Status: Canonical · Open Standard

Effective Date: 17 August 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Simulation Model & Environment

Canonical Definition

Simulation Model & Environment Standard (SMES) defines the governance conditions under which a simulation model and its simulation environment are constructed, structured, configured, parameterized, bounded, versioned, controlled, and maintained as an identifiable execution context capable of supporting governed simulation activity.

It establishes the methodological conditions through which the selected representation of reality and its governed assumptions are transformed into an executable simulation structure while preserving explicit relationships between model components, environmental conditions, boundaries, parameters, configurations, versions, dependencies, and intended simulation purpose.

SMES does not determine whether the simulation is universally valid, whether its outputs justify operational reliance, or whether the simulated environment is equivalent to actual Operational Reality.

It governs the model and environment in which simulation activity will occur.

Purpose

A simulation does not execute directly against reality.

It executes through a constructed model operating within a defined simulation environment.

That model may represent: physical systems, biological systems, economic systems, organizations, infrastructure, software, AI systems, autonomous systems, human behavior, environmental conditions, multi-agent environments, scientific phenomena, operational processes, or combinations of these.

The simulation environment may contain: objects, actors, rules, relationships, parameters, constraints, environmental states, interactions, dependencies, temporal conditions, spatial conditions, stochastic processes, external interfaces, and execution controls.

Without governance over this layer, a simulation may be formally defined yet execute through a model or environment that no longer corresponds to the intended Simulation Object, Purpose, Scope, Reality Representation, or Assumptions.

SMES exists to prevent that separation.

Core Governance Question

How must the simulation model and simulation environment be constructed, configured, bounded, parameterized, versioned, and controlled so that the simulation executes within an identifiable and governable simulation context?

Governance Objectives

SMES establishes governance over: simulation model structure, simulation environment configuration, model-environment relationships, simulation boundaries, boundary interactions, simulation parameterization, model dependencies, environment dependencies, model configuration, environment configuration, model states, environment states, internal interactions, external interfaces, model versioning, environment versioning, configuration continuity, change governance, execution-context identification, reproducibility conditions, and model-environment traceability.

The objective is not to prescribe a specific modeling technology.

The objective is to ensure that whatever model and environment are used remain identifiable, bounded, controlled, reconstructable, and connected to the simulation they are intended to support.

Simulation Model

A Simulation Model is the governed representation through which selected characteristics, relationships, behaviors, mechanisms, states, or dynamics of the Simulation Object or represented reality are made executable within a simulation.

A Simulation Model may consist of: one model, multiple interacting models, hierarchical models, coupled models, mathematical models, statistical models, physics-based models, behavioral models, computational models, AI models, agent-based models, digital twins, hybrid models, or other simulation structures.

The technical form does not determine the governance requirement.

Where the structure materially affects simulation behavior, it must remain governable.

Simulation Environment

A Simulation Environment is the governed execution context within which the Simulation Model operates.

It may define: environmental conditions, interacting entities, spatial structure, temporal structure, available resources, environmental rules, constraints, communication pathways, external stimuli, failure conditions, disturbances, operating conditions, or other contextual elements affecting simulation execution.

The Simulation Environment is not Operational Reality.

It is a constructed simulation space intended to represent selected operational or hypothetical conditions. Model–Environment Distinction

SMES distinguishes:

Simulation Model

from:

Simulation Environment

The model defines the executable representation of the relevant object, system, process, actor, or phenomenon.

The environment defines the conditions within which that representation operates and interacts.

In complex simulations the distinction may be distributed or technically blurred.

The governance distinction must nevertheless remain sufficiently clear to determine:

what belongs to the model,

what belongs to the environment,

and:

how they interact.

Model Structure Governance

The structure of the Simulation Model should remain sufficiently defined to identify: model components, component roles, component relationships, dependencies, interfaces, state transitions, internal logic, relevant constraints, and material model boundaries.

The required depth of structural governance depends upon the Simulation Purpose, Scope, complexity, and consequence.

Model Composition

Where multiple models are combined, the simulation should preserve the composition relationship.

Conceptually:

Model A + Model B + Model C → Composite Simulation Model

The composite model should not obscure materially different: assumptions, representations, parameters, versions, dependencies, or operating conditions

of its constituent models.

Model Dependency

Simulation Models may depend upon: other models, libraries, external services, datasets, parameter sources, environmental models, computational infrastructure, AI systems, physical models, or other simulation components.

Material dependencies should remain identifiable.

Environment Configuration

The Simulation Environment should have an identifiable configuration sufficient to determine the conditions under which simulation execution occurs.

The configuration may include: environment type, environment state, environmental variables, active entities, constraints, interfaces, temporal configuration, spatial configuration, stochastic conditions, environmental rules, and external dependencies.

Execution Context

SMES establishes the Simulation Execution Context as the combined governed state of:

Simulation Model + Simulation Environment + Configuration + Parameters + Boundaries + Versions

This context provides the structural basis for subsequent simulation execution.

Simulation Boundary

The Simulation Boundary determines which interactions are governed as occurring inside the simulation and which originate outside the simulated system.

A boundary may separate:

Simulated Object ↔ Simulated Environment

Simulated Environment ↔ External Input

Model Component ↔ Model Component

or:

Simulation ↔ External System

depending upon the architecture of the simulation.

Boundary Interaction

Defining a boundary is insufficient where information, forces, resources, signals, commands, states, or other effects cross it.

SMES therefore governs Boundary Interactions.

A material Boundary Interaction should establish: what crosses the boundary, from where, to where, under which conditions, through which interface, and with what effect upon the simulation.

Open and Closed Simulation Boundaries

A simulation may operate with:

Closed Boundaries

where relevant interactions are internally represented,

or:

Open Boundaries

where external systems, data, humans, models, services, or environments influence execution.

Open boundaries require sufficient interface governance to preserve execution-context clarity.

External Simulation Interfaces

External interfaces may connect the simulation to: live data, operational systems, external models, human operators, AI systems, sensors, external databases, hardware, cloud services, or other simulation environments.

The existence of an interface should not silently expand the Simulation Scope. Parameterization

A Simulation Parameter is a governed value, range, distribution, rule, state,

or configuration variable that materially influences simulation behavior.

Parameterization should preserve sufficient information to determine:
parameter identity, parameter meaning, applicable unit, value or range, source
or basis where relevant, permitted variation, dependency, applicable model or
environment component, and version.

Parameter Classes

Parameters may include: fixed parameters, variable parameters, stochastic parameters, calibrated parameters, derived parameters, scenario-controlled parameters, environment parameters, model parameters, boundary parameters, and externally supplied parameters.

Different parameter classes may require different governance controls.

Parameter Range

Where a parameter is governed by an applicable range, the simulation should distinguish:

Permitted Range

from:

Executed Value

and where relevant:

Boundary / Out-of-Range Value

A technically executable value does not automatically establish a governed parameter value.

Parameter Dependency

Changing one parameter may affect: other parameters, model behavior, environment state, scenario conditions, execution pathways, or outputs.

Material parameter dependencies should remain identifiable where necessary for reconstruction and change assessment.

Configuration Governance

A simulation configuration is the specific combination of model, environment, parameters, boundaries, interfaces, dependencies, and versions used for a simulation context.

SMES requires material configurations to remain identifiable.

Conceptually:

Model Version + Environment Version + Parameter Set + Boundary Configuration +
Interface Configuration = Simulation Configuration Configuration Identity

Where materially different configurations can produce materially different simulation behavior, each configuration should be distinguishable.

A Simulation Configuration should therefore not be represented as unchanged where material configuration changes have occurred.

Configuration Control

Configuration control should govern: creation, approval where required, modification, activation, deactivation, replacement, comparison, archival, and reconstruction

of material simulation configurations.

Model State

A model may contain internal states that affect behavior.

Where state materially affects execution, the initial model state should be identifiable.

Examples may include: initialized, degraded, failed, partially operational, trained, untrained, calibrated, or another domain-specific state.

Environment State

The Simulation Environment may likewise begin in a defined state.

The initial environmental state may materially influence execution and should therefore be governed where relevant.

Initial Conditions

Initial conditions establish the simulation state at execution start.

They may include: object state, environment state, actor state, system configuration, resource state, position, velocity, time, population, network state, or other relevant conditions.

Initial conditions should remain distinguishable from later simulation-generated states.

Temporal Structure

A Simulation Environment may govern time through: continuous time, discrete time, event-driven time, accelerated time, synchronized time, asynchronous time, historical replay, or another temporal structure.

Where temporal structure materially affects results, it should remain explicit.

Spatial Structure

Where space materially affects the simulation, the environment should define relevant: coordinate systems, dimensions, regions, distances, topology,

movement rules, spatial resolution, or boundaries.

Stochastic Environment

Where randomness is materially used, the simulation environment should preserve sufficient governance over: stochastic process, probability distribution, randomization configuration, seed where relevant, sampling behavior, and affected components.

This supports reproducibility and interpretation of variation across executions.

Deterministic Environment

Where identical governed inputs are expected to produce identical simulation behavior, that expectation should be explicitly established.

Unexpected nondeterminism may then become visible rather than being silently accepted. Model–Environment Compatibility

A Simulation Model should be sufficiently compatible with the Simulation Environment in which it is executed.

Compatibility may concern: interfaces, units, states, timing, resolution, data formats, assumptions, boundary conditions, and interaction rules.

Compatibility should not be presumed solely because technical execution succeeds.

Representation Continuity

The model and environment should remain connected to the governed Reality Representation established under the preceding SIMULOS® lifecycle.

Conceptually:

Selected Reality → Governed Representation → Governed Assumptions → Simulation Model & Environment

SMES does not redefine Reality Representation.

It operationalizes that representation into an executable simulation structure.

Assumption Binding

Material Simulation Assumptions should remain traceably connected to the model or environment elements in which they are implemented.

Conceptually:

Simulation Assumption → Model / Environment Element → Parameter / Rule / Boundary / Interaction

This prevents assumptions from becoming invisible once encoded into the

simulation.

No Hidden Model Substitution Principle

A materially different model must not be substituted for the governed Simulation Model without preserving the change and its consequences for the simulation context.

No Hidden Environment Substitution Principle

A materially different Simulation Environment must not be treated as equivalent merely because the same Simulation Model continues to execute.

No Technical Execution Equivalence Principle

SIMULOS® establishes:

Technical executability does not demonstrate model-environment governance sufficiency.

A model may run successfully while: using the wrong environment, using incompatible parameters, operating outside governed boundaries, relying upon obsolete assumptions, or executing under a materially different configuration.

Version Governance

Material versions of the Simulation Model and Simulation Environment should remain identifiable.

Version governance should distinguish changes capable of affecting: model structure, environment behavior, parameters, interfaces, assumptions, dependencies, scenarios, execution, or output interpretation.

Model Version

A Model Version should identify a materially distinguishable state of the Simulation Model.

The version should remain connected to the simulation executions in which it participated.

Environment Version

An Environment Version should identify a materially distinguishable state of the Simulation Environment.

The same model executed against different environment versions may represent different simulation conditions.

Version Pairing

Where model and environment versions interact materially, the combination should remain identifiable.

Conceptually:

Model v3.2 + Environment v5.1

is not automatically equivalent to:

Model v3.2 + Environment v5.2 Configuration Version

Where parameters, boundaries, interfaces, or other configuration elements materially change without requiring a new model or environment version, a distinct Simulation Configuration Version may be established.

Change Governance

Material model or environment changes should be assessed before they silently propagate into later simulation activity.

Changes may affect: representation, assumptions, parameters, scenarios, inputs, execution, outputs, correlation, or downstream reliance.

SMES governs identification of the change and its simulation-context consequences.

Other SIMULOS® lifecycle spaces govern the affected downstream questions.

Material Change

A Material Model or Environment Change is a change capable of materially altering simulation behavior, execution conditions, representational meaning, scenario behavior, or output interpretation.

Materiality depends upon the Simulation Purpose and Scope.

Change Classification

Changes may be classified as:

Non-Material

Material

Critical

or:

Undetermined

according to the applicable governance context.

Configuration Drift

Simulation Configuration Drift occurs when the active simulation configuration gradually or unintentionally diverges from its governed configuration.

Drift may arise through: parameter changes, dependency updates, environment updates, external interface changes, model updates, infrastructure changes, or undocumented modifications.

SMES requires material drift to remain detectable.

Configuration Lock

Where simulation reproducibility or high-consequence use requires strict control, a configuration may be locked for a defined simulation campaign, test, scenario set, or execution.

A locked configuration should not be altered without governed change.

Configuration Reconstruction

SMES should enable reconstruction of the configuration under which a simulation was executed.

At minimum, where materially relevant, reconstruction should answer:

Which model was used?

Which environment was used?

Which versions were active?

Which parameters were applied?

Which boundaries and interfaces existed?

Which relevant assumptions were bound to the model or environment?

Reproducibility Conditions

SMES does not require every simulation to be perfectly reproducible.

Some simulations are inherently stochastic, adaptive, distributed, or dependent upon changing external conditions.

It requires that the conditions affecting reproducibility be sufficiently governed to determine whether reproduction is:

Expected

Conditionally Expected

Not Expected

or:

Undetermined Model–Environment Traceability

Material simulation structures should support traceability across:

Simulation Object



Simulation Purpose & Scope



Reality Representation



Simulation Assumptions



Simulation Model



Simulation Environment



Configuration



Parameters



Version



Execution Context

This chain creates the governed structural basis for later simulation execution.

Model & Environment Governance Record

A governed SMES implementation may preserve: Simulation Model Identifier, Simulation Environment Identifier, model description, environment description, model structure, environment configuration, model dependencies, environment dependencies, boundary definitions, boundary interactions, external

interfaces, parameter set, parameter dependencies, initial conditions, model state, environment state, temporal structure, spatial structure, stochastic configuration, model version, environment version, configuration version, assumption bindings, representation references, change history, configuration status, drift status, and complete traceability references.

Governance Flow

SMES establishes the following governed progression:

Governed Reality Representation

↓

Governed Simulation Assumptions

↓

Simulation Model Structure

↓

Simulation Environment Configuration

↓

Boundary & Interaction Definition

↓

Parameterization

↓

Model–Environment Compatibility

↓

Configuration Establishment

↓

Version Binding

↓

Controlled Simulation Execution Context

This flow transforms the governed conceptual representation of the simulation into a controlled model-environment structure capable of supporting subsequent

simulation inputs, scenarios, and execution.

Governance Boundary

SMES begins when a sufficiently governed:

Simulation Object + Simulation Purpose & Scope + Reality Representation +
Simulation Assumption Basis

must be transformed into an executable simulation structure.

SMES ends when the:

Simulation Model + Simulation Environment + Boundaries + Parameters +
Configuration + Versions

have been sufficiently established and controlled to support governed
downstream simulation activity.

SMES does not govern: actual Operational Reality, general integrity
governance, universal model validation, general Validation Method governance,
simulation input fitness, scenario design, simulation execution, simulation
output governance, reality correlation, or operational reliance.

These responsibilities belong to adjacent SIMULOS® spaces or complementary
OOF® architectures.

Minimum Governance Requirements

A conforming implementation of SMES should establish:

1. an identifiable Simulation Model, 2. an identifiable Simulation Environment, 3. sufficiently defined model structure, 4. sufficiently defined environment configuration, 5. explicit model-environment relationships, 6. governed simulation boundaries, 7. material Boundary Interactions, 8. governed parameterization, 9. material parameter dependencies, 10. identifiable initial conditions, 11. material model and environment dependencies, 12. model-environment compatibility, 13. assumption-to-model/environment binding where material, 14. model and environment version identification, 15. configuration identification, 16. material change governance, 17. configuration drift visibility, 18. reconstruction capability, 19. and complete model-environment traceability.

Governance Outputs

SMES may produce: Simulation Model Identifier, Simulation Environment Identifier, Simulation Model Structure Record, Simulation Environment Configuration Record, Model–Environment Relationship Map, Simulation Boundary Record, Boundary Interaction Record, External Simulation Interface Record, Simulation Parameter Register, Parameter Dependency Map, Initial Condition Record, Model State Record, Environment State Record, Model Dependency Record, Environment Dependency Record, Model–Environment Compatibility Record,

Assumption Binding Map, Simulation Model Version Record, Simulation Environment Version Record, Simulation Configuration Record, Simulation Configuration Version, Simulation Configuration Change Record, Simulation Configuration Drift Record, Simulation Configuration Lock Record, Simulation Execution Context, Model–Environment Traceability Map, and Model & Environment Governance Record.

Relationship to Other OOF® Architectures

SMES operates within SIMULOS® — Simulation Governance Architecture and remains bounded from adjacent OOF® governance spaces.

ORA™ governs actual Operational Reality and its operational conditions. SMES governs the constructed simulation environment in which selected aspects of reality are represented for simulation.

VALIDOS® governs whether a model, simulation, output, claim, or related object satisfies an applicable validation inquiry. SMES establishes the simulation-specific model and environment governance basis but does not itself declare universal validity.

INTEGROS® may govern integrity of models, configurations, parameters, versions, records, and related objects. SMES governs their simulation-specific structural and environmental role.

ADIT® may govern audit evidence generated from model, environment, configuration, change, and execution activity.

AGA™ may govern accountability for model and environment decisions, changes, approvals, and control responsibilities.

Other OOF® architectures may provide specialized governance conditions consumed by the simulation where their governed objects are relevant.

SMES does not replace those architectures.

It governs specifically how the model and environment through which a simulation will operate are established and controlled.

Architectural Position

Simulation Model & Environment is the fifth Parent Standard of SIMULOS® — Simulation Governance Architecture.

It follows:

Simulation Object → Simulation Purpose & Scope → Reality Representation → Simulation Assumptions

and establishes the structural bridge into executable simulation activity.

The lifecycle continues from SMES into governance of:

Simulation Inputs & Data → Scenario Design → Simulation Execution → Simulation Output → Reality Correlation → Operational Reliance → Drift & Resimulation

SMES therefore occupies a critical architectural position:

Representation becomes executable structure here.

Foundational Principle

A simulation becomes structurally governable only when the model and environment through which it operates are identifiable, bounded, configured, parameterized, versioned, controlled, and traceably connected to the reality representation and assumptions they are intended to implement.

Canonical Closing Statement

Simulation Model & Environment Standard establishes the structural execution foundation of SIMULOS® by governing how Simulation Models and Simulation Environments are constructed, configured, bounded, parameterized, versioned, controlled, changed, and reconstructed. By preserving explicit relationships between model structure, environment configuration, boundary interactions, parameters, dependencies, initial conditions, versions, governed assumptions, and Reality Representation, SMES ensures that simulation execution cannot become detached from the simulation context it is intended to represent and provides the controlled model-environment basis required for subsequent input, scenario, execution, output, correlation, and lifecycle governance.

Module Architecture

→ [Simulation Model Structure Module \(SMMM\)](#)

→ [Simulation Environment Configuration Module \(SECM\)](#)

→ [Simulation Boundary Interaction Module \(SBIM\)](#)

→ [Simulation Parameterization Module \(SPHM\)](#)

→ [Simulation Environment Versioning Module \(SEVM\)](#)

Related Documents

→ [About Simulation Model & Environment Standard \(SMES\)](#)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>